

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks									
Q.1 Solve Any Two of the following.		12									
A) What is sharpening special filter, describe with the help of Laplacian filter?	CO1	6									
B) Write a Short Note on: a) Contrast stretching. b) Adaptive Histogram Equalization (AHE).	CO1	6									
C) 3) Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using neighborhood using (3x3) all the filters below: 1) Box filter, 2) Weighted average filter, 2) 3) Median filter, 3) 4) Min filter, 5) Max filter.	CO1	6									
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6	0	9									
8	2	7									
1	4	3									
Q.2 Solve Any Two of the following.		12									
A) Explain the concept of image transform	CO2	6									
B) Write a short note on: • Walsh-Hadamard Transform • Slant Transform • Haar Transform • DCT	CO2	6									
C) Differentiate between DFT (Discrete Fourier Transform) and DCT (Discrete Cosine Transform) in image processing.	CO2	6									
Q.3 Solve Any Two of the following.		12									
A) Explain the region-based segmentation, including region growing, region splitting and merging concept?	CO3	6									
B) Explain the concept of Prewitt, Sobel, and Canny edge detection methods.	CO3	6									
C) Describe the differences between region-based segmentation and edge-based segmentation.	CO3	6									

Q.4 Solve Any Two of the following.		12
A) What are the main characteristics that make BRISK a binary feature extraction method	CO4	6
B) What is SURF, and what are the main advantages of SURF over SIFT?	CO4	6
C) The Bag-of-Words model is often employed to represent features in a dataset. Could you describe how this model works and the steps involved in transforming raw visual data into feature vectors using Bag-of-Words?	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Explain the multi-stage architecture of Fast R-CNN and how it improves upon R-CNN.	CO5	6
B) Explain object detection by YOLO.	CO5	6
C) Describe the role of a generator and a discriminator in a GAN.	CO5	6

***** End *****